

Ecological Predictive Cruise Control of Connected Electric Vehicle with Predecessor Velocity Prediction and Road Grade Preview

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Abstract—This paper proposes a predictive adaptive cruise control (PACC) strategy for connected electric vehicle (EV) which aims to minimize vehicle energy consumption with safe car-following. First, a Long Short-Term Memory (LSTM) is designed to predict the velocity of preceding vehicle. Second, the safe car-following spacing is derived by using the theory of state invariant set. Finally, an integrated PACC controller based on nonlinear model predictive control (NMPC) is proposed to achieve energy-efficient and safe vehicle cruising control. Simulation results show that the proposed PACC strategy can improve vehicle energy efficiency while maintaining a safe distance with preceding vehicles.

Keywords—Adaptive Cruise Control, Eco-driving, Electric Vehicle

I. INTRODUCTION

The recent advancements in artificial intelligence (AI) and sensing technology promote the development of Advanced Driving Assistance Systems (ADAS) [1]. As one of the earliest ADAS applications, the adaptive cruise control (ACC) is benefited from the emerging of AI and vehicular wireless communication techniques. For example, the neural networks aids us with brain to think further, while vehicle-to-everything (V2X) endow us with eyes to see further [2].

ACC is developed from Cruise Control (CC), freeing drivers from stepping on a pedal to maintain the vehicle at a constant speed by automatically adjusting the throttle. In addition to CC, ACC takes preceding vehicle into consideration, utilizing sensors like radar to estimate relative distance from the predecessor to avoid collision [3]. The core of ACC is its gap policy, which has been extensively studied since at least 1994[4]. It is plainly to come up with the idea of fixed distance policy which is not robust to the changing speed of vehicles [5]. Another commonly used strategy is called Constant Time Gap (CTG) policy that the desired relative distance is proportional to the leader's velocity, follows as $d_{des} = l_{min} + \tau v_{pre}$, where l_{min} is the minimum relative distance for static situation, τ denotes the time gap and v_{pre} is the predecessor's speed [6]. While such method still cannot deal with the various conditions including weather, roads, and vehicle dynamics.

The aforementioned two policies are classical but flawed, for which extensive research in ACC is still put in. Some researchers try to imitate human drivers' maneuver with a mathematical model, by which makes passengers feel more

comfortable and 'natural' [7]. However, human drivers are not flawless for demanding a reaction time, especially under emergency circumstances it may lead to a sudden brake which is parlous to the passengers intact being [8]. Some researchers introduce the idea of field theory and use an artificial potential field to assess the risk of collision, yet it may fall into a local minimum [9]. A control invariant set is used in [10] to find the bounds of acceleration for a platoon, which is truly safe but somehow conservative. Above all these methods, the main contradiction lies between the desire for safety and the unknown, potentially dangerous, exogenous states. Thus, it is an approach to know and utilize the future knowledge in advance that eliminate the dilemma of innocence and curiosity.

Fortunately, via V2I communication and GPS/GIS, we can foresee the future road topography and speed limit, which can be integrated with optimization problem formulation, including objectives and constraints. Such cruise control is known as Predictive Cruise Control (PCC), with merits of saving energy, cutting emission, and facilitating efficiency [11]. Besides, with progressive development of Connected Autonomous Vehicles (CAV), V2V technique enables the planned future trajectories transmitted to adjacent vehicles for distributed motion control [12]. In fact, the penetration rate of CAV is still low, which reflects the truth that there is a greater possibility that a CAV is following a human-driven vehicle [13]. Hence, it is still worthy to study the driver behavior prediction.

Among the recent vehicle motion prediction techniques, recurrent neural network (RNN) has drawn much attention compared to those probabilistic method like hidden Markov Chains, Bayesian Network or Kalman Filter [14][15]. Due to the property of speed trajectory, a time-dependent series type data, RNN is able to relate the past time steps to present, capture temporal features from the past and recent time-series data and predict driver's future motion. Two of the most famous RNN implementations are gated recurrent unit (GRU) and long short-term memory (LSTM), which have similar accuracy but a slight difference on performance [16].

In this paper, a predictive ACC (PACC) strategy based on nonlinear model predictive control (NMPC) is proposed to improve vehicle energy efficiency while maintaining safety. The concept of control invariant set is applied to create a safe state constraint set for generating safe relative distance boundaries. The historical trajectory of preceding vehicle is accessible via V2X network to train and update a LSTM neural network for leader velocity prediction. The upcoming road slope and speed limits knowledge are available via V2X network and GPS/GIS for the utilization for both safe state

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constraints set and LSTM-based leader predictor to facilitate security and economy. The proposed PACC structure is illustrated in Fig.1.

The remainder of this paper is as follows. The vehicle longitudinal dynamics and energy consumption are modeled in Section II. Section III introduced the training and verification of LSTM-based leader velocity predictor. Safe state constraint set with control invariant set and NMPC problem formulation are formulated in Section IV. In Section V, simulations on safety and energy economy are presented. Finally, Section VI concludes the paper with reflection and perspective.

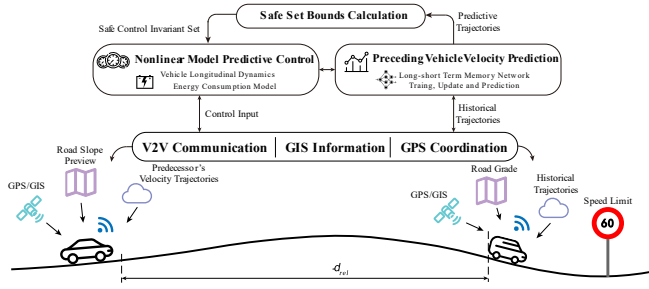


Fig. 1. Structure of Predictive Adaptive Cruise Control System

II. VEHICLE DYNAMICS MODELING

In this section, we will present the longitudinal dynamic model and energy consumption model of ego EV, and a combined leader-follower pair model that leads to the state constraint set utilizing the idea of control invariant set to guarantee the safe relative distance between ego vehicle and preceding vehicle. The symbol in this section is used to indicate measurements throughout the paper.

A. Vehicle Longitudinal Dynamic Model

In this paper, we consider an EV with two in-wheel motors that is capable of propel the vehicle and brake with regenerative power recovery. Since in a car-following scenario, we only consider the longitudinal motion and dynamics of the vehicle. The longitudinal forces applied to the vehicle can be expressed as:

$$m\dot{v} = \underbrace{\frac{\Sigma T}{R}}_{\text{Torque}} - \underbrace{\frac{1}{2}C_D A_f \rho v^2}_{\text{Air Drag}} - \underbrace{mg \sin \theta}_{\text{Gradient}} - \underbrace{fmg \cos \theta}_{\text{Rolling}} \quad (1)$$

where T represents the torque on a single wheel, R is the radius of the wheel. The left side of the equation equals to the propulsion force, which provide the power to accelerate the vehicle. And on the right side is the resistance forces, including aerodynamic resistance, rolling resistance, and road gradient resistance. The first term in the right side of the above formula is aerodynamic resistance, which is determined by air drag coefficient C_D , frontal surface A_f , air density ρ , and longitudinal velocity v . The second term is road gradient resistance, mainly decided by vehicle mass m , gravitational acceleration g , road slope θ . While the last term rolling resistance is controlled by vehicle mass, road slope, gravitational acceleration and roll coefficient f .

B. Energy Consumption Model

Assuming that the host vehicle is powered by two in-wheel motors. A quasi-static model is utilized to describe the output torque and electricity consumption of the vehicle.

$$P_{motor} = T_{motor} \omega_{motor} \eta_{motor}^{\text{sign}(T\omega)} \quad (2)$$

where T_{motor} and ω_{motor} is the torque output and rotational speed of in-wheel motor, η_{motor} is the coefficient representing the original efficiency of motor itself.

The η_{motor} can be calculated by linear interpolation on a characteristics curve given in Fig.2. $\text{sign}(T\omega)$ is the sign function that tell if the product of torque and rotational speed is positive or negative. If T and ω have the same direction, the motor is propelling the vehicle no matter forward or backward, and the motor is transmitting the energy from battery to wheels. While if they have opposite directions, the motor is braking, operating as a generator, so the energy flows from wheels as kinetic energy to the battery as electricity.

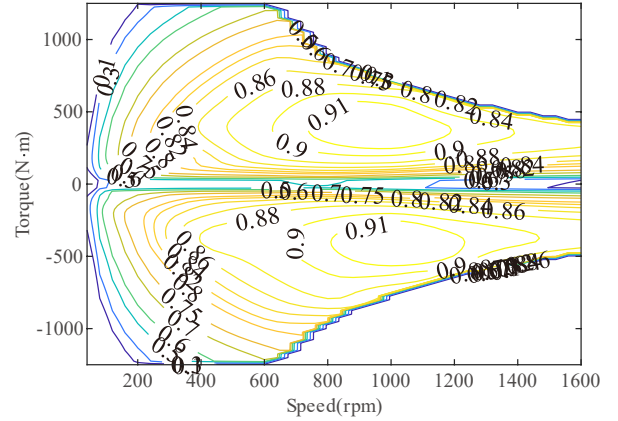


Fig. 2. Efficiency MAP of in-wheel motor

In this case, we assume that the power demand is split evenly to both motors. Hence, both motors are powered by battery, whose power follow the equation as:

$$P_b = 2P_{motor} \eta_e^{\text{sign}(T\omega)} \quad (3)$$

where P_b is the battery power, η_e is the efficiency coefficient of power converter circuit. Function $\text{sign}(T\omega)$ has the same meaning that has been talked before.

The battery model is derived from equivalent circuit model (as shown in Fig.3), and the changing rate of state of charge (SOC) level can be deduced as following equation as:

$$\dot{SOC} = -\frac{I_b}{E_b} = -\frac{V_{oc} - \sqrt{V_{oc}^2 - 4P_b R_b}}{2E_b R_b} \quad (4)$$

where I_b is the current of battery, E_b is the capacity, V_{oc} is open circuit voltage, R_b is the internal resistance.

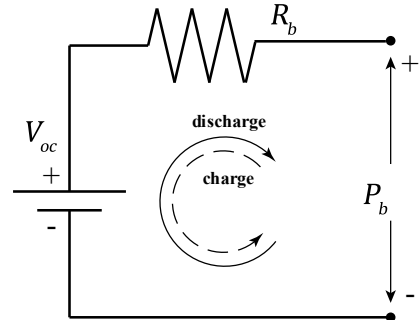


Fig. 3. Equivalent Circuit of Battery

III. PREDECESSOR VELOCITY PREDICTION

In this case, we assume that the preceding and ego vehicles are both connected, for which the current status of leader can be transmitted to the follower, but for the future speed trajectory plan considering the leader could be human-driven. The predicted velocity time series is integrated into the prediction horizon of MPC to optimize the inter-vehicle distance.

A. LSTM Network Establishment

In the foreseeable future, the market share of CAV is still far from ideal, i.e., there is a big possibility that a CAV follows a Human-driven vehicle. Hence, it is of much benefit and significance to predict the motion of leader vehicle. As a typical type of RNN, long short-term memory network has been proven to be promising in time series prediction. The key component of LSTM network (as shown in Fig.4) is its hidden layer, also known as Memory Cell, comprising input gate, forget gate, output gate that enabling the cell to determine which part of the input sequence will be stored, which part of the old memory will be forgotten and how much of the memory will be extracted to perform the prediction. Such design could well deal with the vanishing gradient problem and better capture long range dependencies within sequences.

In this paper, we aim to predict the preceding vehicle's longitudinal motion in a short-term horizon. Thus, a NGSIM I-80 dataset is utilized to train the network for its well describing the dynamic characteristics of passenger vehicles. The input features are vehicle speed, acceleration, position, and current road grade sequence, along with the next time step's velocity sequence as output feature. The output prediction sequence length is 20 steps, while input sequence dimension is 100-time steps. Once a prediction is made at time step k , the input sequence window and prediction window move on one step simultaneously. To elevate the prediction accuracy and training efficiency, hidden layers are carefully chosen to be 200 and max training epochs are 400. The detailed LSTM RNN establishment is shown in TABLE.1.

TABLE I. LSTM NETWORK INPUTS AND OUTPUTS

LSTM Layers	Variables	Unit	Layers Number
Input Features	Leader Velocity	m/s	100
	Leader Acceleration	m/s ²	
	Leader Position	m	

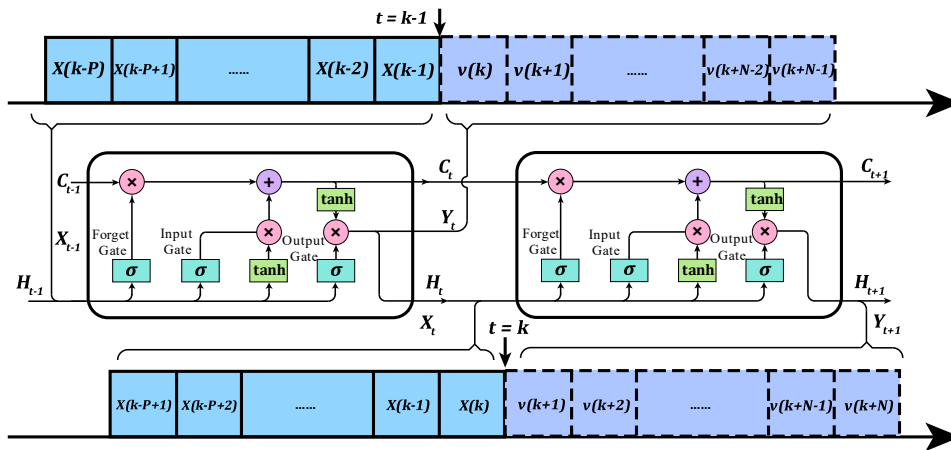


Fig. 4. LSTM Cell Prediction Structure and Flowchart

	Road Grade	rad	
Output Features	Leader Velocity	m/s	20

Since input features differ in magnitudes immensely, standardizing the input data is required in case small values are annihilated or training falling into divergence. The standardization process, aligning the data to zero mean and unit variances, should be applied to forecast applications in the same way. The standardization follows the Equation (9.a-c).

$$\bar{X} = [\bar{s} \quad \bar{v} \quad \bar{a} \quad \bar{\theta}]^T = \frac{1}{N} \sum_{k=1}^N X_i(k) \quad (9.a)$$

$$sig = \sqrt{\frac{1}{n-1} \sum_{k=1}^N (X_i - \bar{X}_i)^2} \quad (9.b)$$

$$X^* = \frac{X_i - \bar{X}_i}{S} \quad (9.c)$$

where X and $\bar{X}$ are the raw data vector containing the historical status values of preceding vehicle and its mean value vector, sig is the standard deviation for each variant and X^* is the disposed vector transmitted to the input feature.

B. Verification of LSTM Performanc

Before we test the performance of proposed LSTM network, a general evaluation criterion, known as Root Mean Square Error, is introduced first to assess the accuracy between LSTM prediction and observed values.

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (Y(t) - \hat{Y}(t))^2} \quad (10)$$

where $Y(t)$ is observed actual velocity at time step t and $\hat{Y}(t)$ is the predicted one. To see if the trained network is valid and capable to make logical predictions, we take the first 90% part of the training data to train the LSTM neural network, and the 10% left to verify the prediction accuracy. Due to the size of the input training data, the training data is split into several batches. The prediction result of the trained network is shown in Fig.5(a), as shown in the picture that, the prediction curve basically follows the form of the observation values of the last part of the dataset, and the RSME is smaller than 0.05.

After that, the generalization performance of the trained network when applied to other driving cycles are compared. The HWFET driving cycles and US30 driving cycles are chosen. Due to the size of data and limited computation power, typical section of curves with changing velocity are chosen to predict and the results are shown in Fig.5(b) and Fig.5(c), respectively. The RMSE of them are both smaller than 0.1, which is acceptable considering the sampling time interval and different driving characters of datasets.

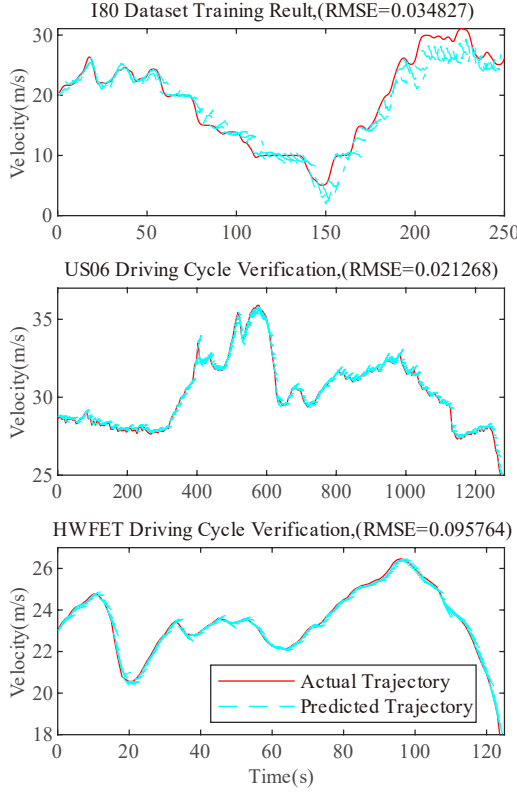


Fig. 5. (a). Training Result of one section of NGSIM(I-80) dataset; (b). Verification on one section of US06 Driving Cycle; (c) Verification on one section of HWFET Driving Cycle

IV. PREDICTIVE ADAPTIVE CRUISING CONTROL DESIGN

In this section, the proposed NMPC-based predictive controller is presented, along with the LSTM-based leader velocity predictor. In this case, we assume that the preceding and ego vehicles are both connected, for which the present status of leader can be transmitted to the follower, but for the future speed trajectory plan considering the leader could be human-driven. The predicted velocity time series is integrated into the prediction horizon of MPC to optimize the inter-vehicle distance. The NMPC problem is solved by inner-point method.

A. State Constraint Set

The most essential function of ACC is security, i.e., keeping a safe distance away from the preceding vehicle regardless the motion the leader takes. Obviously, larger the gap is, safer the follower will be. However, large gap may deviate traffic efficiency and potentially increase air drag. Thus, regulating a sensible gap between leader and follower is of much significance to ACC. Considering a two-vehicles

combined ACC problem that committed to find a minimum safe distance which is dull enough to the leader's most urgent brake.

$$\begin{bmatrix} \Delta \dot{s} \\ \dot{v}_{ego} \\ \dot{\hat{v}}_{lead} \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta s \\ v_{ego} \\ \hat{v}_{lead} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} g(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \hat{h}(t) \quad (11)$$

where Δs is the relative distance between preceding vehicle and ego vehicle, v_{ego} and $\hat{v}_{lead}$ are the host car's velocity and preceding car's velocity which is predicted in this case. $g(t)$ is control variant, which describes the longitudinal acceleration of ego car, while preceding vehicle's acceleration $\hat{h}(t)$ is seen as disturbance.

Hence, a control invariant set is introduced to provide the safe relative distance bounds. All permissible states of the above system are defined as an admissible set:

$$\mathcal{X} = \{ [\Delta s \ v_{ego} \ \hat{v}_{lead}]^T : \Delta s > l_{min}, v_{ego} \geq 0, \hat{v}_{lead} \geq 0 \} \quad (12)$$

where l_{min} is the minimum headway between ego vehicle and leading vehicle, and in here is the length of leading vehicle. In order to ensure the absence of collision for the above system (11), a safety state constraint set $\Psi \subseteq \mathcal{X}$ is defined as:

$$\Psi = \{ [\Delta s \ v_{ego} \ \hat{v}_{lead}]^T : \Delta s > l_{min}, \Delta s - \frac{\hat{v}_{lead}^2}{2 \hat{h}(t)_{min}} + \frac{v_{ego}^2}{2 g(t)_{min}} \geq l_{min}, v_{ego} \geq 0, \hat{v}_{lead} \geq 0 \} \quad (13)$$

where $\hat{h}(t)_{min}$ and $g(t)_{min}$ denote the minimum acceleration of leader and follower, which can also be seen as their maximum braking acceleration. According to theorem of robust control invariant sets, it can be proven that:

$$\forall x(t_0) \in \Psi, \hat{h}(t) \in \mathcal{H}, \exists g(t) \in \mathcal{G}: x(t) \in \Psi, t > t_0 \quad (14)$$

where $\mathcal{H}$ is the leader's possible acceleration set, $\mathcal{G}$ is the ego vehicle's feasible acceleration set. The above definition can be interpreted as, for any leader's motion, if current state of system (11) lies within the control invariant set, an input of ego can be found to maintain the system safe in any future times. The detailed proof of its being invariant set could be referred to [10].

There are several approaches to determine the boundary of Ψ , most of which are conservative and sufficient. To ensure a satisfied initial state, usually the minimum acceleration and speed should be obtained or estimated, which lays a huge burden over the computation process, let alone the rarity such occasion that needs minimum acceleration and maximum braking force would occur. In light to this, a more practical solution is in need. Owing to the advancement of V2X network and high-resolution GPS/GIS, future road grade and speed limit can be a priori-information for safe set Ψ calculation. And with the help of preceding vehicle predictor mentioned hereinafter, the estimated velocity trajectory will also be embedded to boost the safe set calculation. As mentioned before, our goal is to design a safe set regardless the leader's urgent braking maneuver, hence a Euler-based discrete algorithm is introduced to access the worst situation.

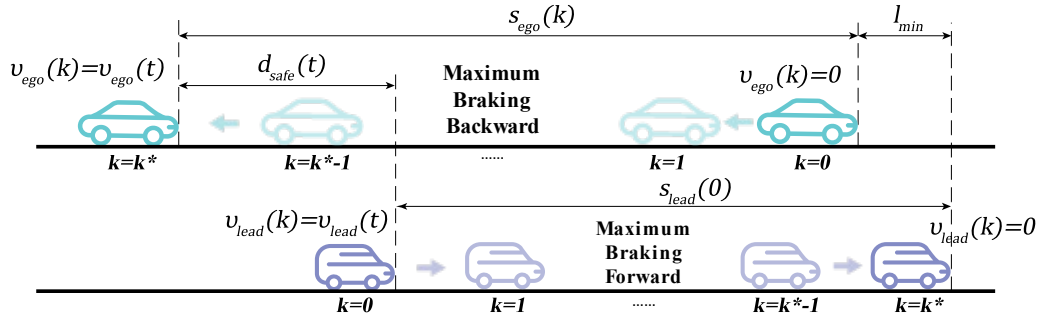


Fig. 6. Illustration of Algorithm 1

Algorithm 1 Algorithm for computing safe invariant set bounds

Input: $v_{lead}(t), v_{ego}(t), s_{lead}(t)$
Output: $d_{safe}(t)$
1: $v_{lead}(0|t) \leftarrow v_{lead}(t)$
2: $v_{ego}(0|t) \leftarrow v_{ego}(t)$
3: $s_{lead}(0|t) \leftarrow s_{lead}(t)$
4: **while** $v_{lead} > 0$
5: $v_{lead}(k+1|t) \leftarrow v_{lead}(k|t) + dt \cdot g(v_{lead}(k|t), \theta(k|t), t)_{min}$
6: $s_{lead}(k+1|t) \leftarrow s_{lead}(k|t) + dt \cdot v_{lead}(k|t)$
7: $s_{ego}(0) = s_{lead}(k) \leftarrow l_{min}$
8: **while** $v_{ego} > 0$
9: $v_{ego}(k+1|t) \leftarrow v_{ego}(k|t) - dt \cdot h(v_{ego}(k|t), \theta(k|t), t)_{min}$
10: $s_{ego}(k+1|t) \leftarrow s_{ego}(k|t) - dt \cdot v_{ego}(k|t)$
11: $d_{safe}(t) = s_{lead}(0) - s_{ego}(0)$

To illustrate the Algorithm 1 concretely, assuming that at time step $k = 0$, the leader decelerated with maximum braking force and stopped at time step $k = k^*$, where we note its final position as $x_{lead}(k^*)$. Then, assuming that ego vehicle accelerated from a constant distance behind the final position of leader with l_{min} , and reached at its current speed at position $x_{ego}(k^*)$, and the safe distance now could be seen as the gap between $x_{ego}(k^*)$ and $x_{lead}(0)$. The main idea of Algorithm 1 is shown in Fig.6.

Thus, the boundary of the safe state constraint set $d_{safe}(t)$ is determined by the above algorithm depending on leader and follower's current velocity and road grade at their current position. Such numerical method will be embedded in the MPC framework talked hereinafter.

B. Optimization Problem Formulation

To integrate into NMPC framework, the vehicle longitudinal dynamic system is discretized temporally by Forward-Euler Method are presented as:

$$x(t+1) = f(u(t), x(t), \theta(t), t) \quad (15)$$

In this case, the system state parameters are $x = [v(t), SOC(t)]$, and control input is the torque $u = [T]$. The primary goal of ACC controller is to refrain the ego vehicle from collision and optimizing the electricity consumption of battery. The NMPC controller solve the following constrained nonlinear optimization problem repeatedly.

$$\begin{aligned} \min_u & J \\ &= \sum_{k=1}^N R_1 \|\delta d(k|t)\|^2 + R_2 \|\Delta a(k|t)\|^2 \\ &+ R_3 \text{sign}(\Delta SOC) \|\Delta SOC(k|t)\|^2 \\ \text{s.t.} & \end{aligned} \quad (16.a)$$

$$x(k+1|t) = f(u(k|t), x(k|t), \theta(k|t), k) \quad (16.b)$$

$$T_{min} \omega(k|t) \leq T(k|t) \leq T_{max} \omega(k|t) \quad (16.c)$$

$$v_{min} \leq v(k|t) \leq v_{max} \quad (16.d)$$

$$SOC_{min} \leq SOC(k|t) \leq SOC_{max} \quad (16.e)$$

$$\begin{aligned} d_{rel}(v(k|t), \hat{v}_{leader}(k|t)) \\ \geq d_{safe}(v(k|t), \hat{v}_{leader}(k|t), \theta(k|t)) \\ + \tau_{comf} v(k|t) + \delta d(k|t) \end{aligned} \quad (16.f)$$

$$\text{where } (\tau_{comf} v(k|t) + \delta d(k|t)) \geq 0$$

where the control input and state variables during step k at current time t is denoted as $u(k|t) = [T(k|t)]$ and $x(k|t) = [v(k|t) SOC(k|t)]$ in this case. ΔSOC denotes the discrete form of SOC difference, and $a(k|t)$ is the acceleration of the vehicle to avoid sudden jerk. N is the length of prediction horizon, and $R_{1 \rightarrow 3}$ are the weighting factors representing the trade-off between different objectives in cost function (16.a), in which the three terms representing car-following capability, driving comfort and energy-saving capability, respectively. Noted that, the energy-saving weighting factor is larger than other two factors by an order of magnitude because ΔSOC has a quite small magnitude. The input and state constraints are limited by inequations (16.c), (16.d) and (16.e). d_{rel} is the relative distance between leader vehicle and ego car which is a function of v and $\hat{v}_{leader}$, while d_{safe} is the safe set boundary derived from the Algorithm 1 aforementioned. To assure the ego vehicle follows the preceding vehicle instead of losing it, we add a soft constraint (16.f) to confine the relative distance within a proper region. In constraint (16.f), the relative distance should be smaller than a comfort distance, which is based on the safe distance above, plus a time-gap τ_{comf} as a comfortable following distance zone. $\delta d(t)$ is the slack variable in constraint (16.f), which is also added to the cost function to minimize the error between actual relative distance and desired comfortable following distance.

The prediction horizon in this case is set as 20 steps, and control horizon is 1, meaning that the head of control input sequence $u(1|t)$ will be applied to the real system. Considering the feasibility of the MPC control problem, the last element of control input series is set to be the minimum of the acceptable control variable $u_{min}(N|t)$, or in another way, the minimum practicable torque $T_{min}(N|t)$ with motor rotational speed $\omega(N|t)$ at time step N .

V. SIMULATION AND DISCUSSION

The vehicle longitudinal dynamic model and proposed PCC controller is established in MATLAB. The vehicle model parameters and NMPC parameters are listed in TABLE II.

To test the safety performance and energy-saving potentials of the proposed PACC, we set up two simulation scenarios with different predecessor velocity profiles, a sinusoidal wave velocity trajectory and a typical part of HWFET driving cycle velocity trajectory, for the PACC installed on ego vehicle to test safety car-following ability and energy-saving ability, respectively. To show the improvement of energy saving performance of proposed PACC controller, we compare it with a constant gap policy ACC controller with a 5m spatial gap.

TABLE II. VEHICLE MODEL PARAMETERS

Symbol	Parameters	Value	Unit
m	Vehicle mass	2500	kg
A_f	Frontal surface	2.5	m ²
ρ	Air density	1.206	/
C_d	Air drag coefficient	0.28	/
f	Rolling friction coefficient	0.015	/
R	Wheel radius	0.36	m
E_b	Energy capacity	48	kWh
V_{oc}	Battery voltage	360	V
R_b	Battery inner resistance	0.012	Ω
T_s	Sampling time step	0.5	s
N	Prediction horizon	20	/
R_1	Car-following weighting factor	1	/
R_2	Comfort weighting factor	10	/
R_3	Energy-saving weighting factor	-10 ⁹	/

A. Safety Constraint performance

A sinusoidal-wave-like velocity profile is applied to the preceding vehicle in the first scenario, with another sinusoidal wave shaped road grade profile. The initial speed of follower is 14.5m/s, and initial position is 7m away from the leader.

In Fig.7(a), the LSTM network can predict the changing speed very well, with little error at turning area, and the proposed PACC attempts to adapt to the oscillatory predecessor's velocity. Fig.7(b) shows that our strategy drives the ego vehicle to follow the leader in case of losing it, meanwhile it keeps a safe gap away from the predecessor instead of crashing into it.

B. Energy-saving performance

In the second case, the preceding vehicle's velocity profile is following a part of the HWFET driving cycle considering real world fidelity. The road slope profile is derived from OpenStreetMap on a highway with similar road length. The initial position of ego vehicle is 5m behind the preceding vehicle, and initial speed of ego vehicle is 12m/s.

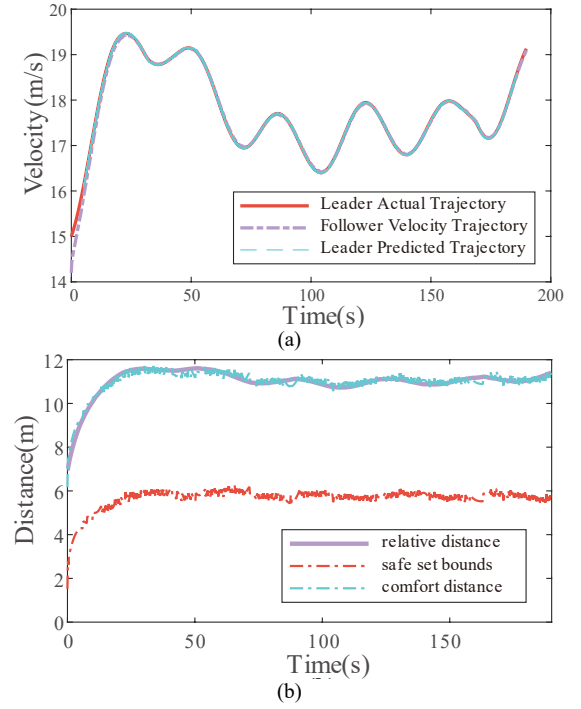


Fig. 7. Simulation results of sinusoidal preceding vehicle (a) predicted and actual velocity trajectory; (b) car-following distances

Fig.8 shows the simulation results. In Fig.8(a), the predicted velocity segments lean to the actual velocity curve, and the leader's velocity and follower's velocity move in tandem and closely. While at the turning area the prediction still has a chance to 'fly off'. It should be noted that, the prediction begins after 20 steps for data collection and neural network update, which is 10s in this case. In Fig.8(b), the relative distance between ego vehicle and preceding vehicle is above the safety bounds constantly, indicating that the proposed policy can make sure of the driving safety, and the relative distance is close to the comfort distance, but smoother than the comfort distance. However, the relative distance under the control of traditional CG-ACC nearly remains the same but keeps oscillate due to the constant gap policy.

Fig.8 (c) and (d) show the control and SOC trajectory for PACC and CG-ACC. As observed, the control input of CG goes up and down to make the ego vehicle keep a constant car-following distance with its preceding one, while the PACC has a smooth control input curve. As a result, the CG consumes more electricity than PACC. Specifically, the PACC improves energy efficiency by 36.8% compared to CG strategy. In other words, traditional ACC under CG mode is easily disturbed by the varying preceding vehicle speed, leading to an oscillation of control variables, which may lead redundant energy consumption. However, the control input of PACC is steadier and smoother, which is beneficial to passenger's driving comfort and vehicle energy economy. The prediction of leader speed is able to avoid sudden changes in control input in some extent.

VI. CONCLUSION

This paper presents a predictive cruise controller utilizing Nonlinear Model Predictive Control Framework and road grade preview information via V2V and GIS system to improve safety, energy saving and efficiency. The preceding vehicle's velocity is collected for training a long short-term memory network for future horizon prediction. To maintain a

considerable and safe gap against the predecessor, the idea of control invariant set is used to create a safe constraint set. The simulation results show the robustness of the proposed PACC strategy against the disturbance of leading vehicle's velocity, and the energy consumption performance is also better than the traditional ACC strategy.

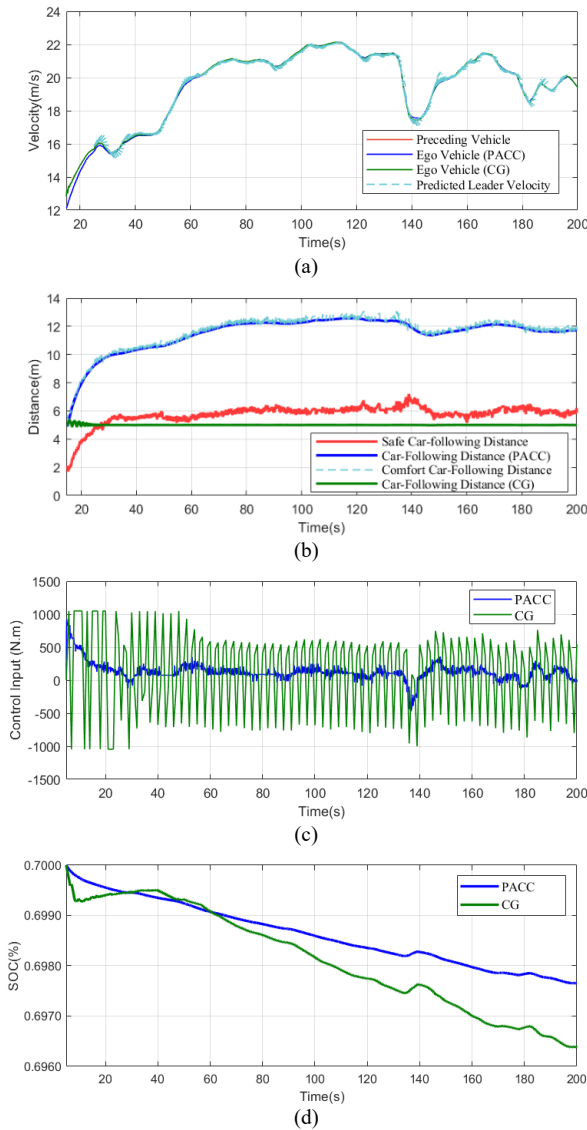


Fig. 8. Simulation results of preceding vehicle driving in HWFET driving cycle (a) velocity profiles of preceding vehicle, and ego vehicle under different strategies; (b) car-following distance; (c) control input profile; (d) SOC trajectory

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